

Run 02 | OME

Global (including India)

IV only

Target Opportunity-motivated entrepreneurship; binary 1 vs. 0

Sample 1,342,884 respondents **Coverage** 89 countries; 2015–2022

Positive cases 100,087 (7.45%) **Case time** 24m 53s

Train / val. / test 913,163 / 161,145 / 268,576 (68% / 12% / 20%, stratified)

Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.

Controls (0): None. Country and respondent ID are excluded as predictors.

Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: XGBoost (0.9714). Highest F1: XGBoost (0.6738). The all-negative accuracy baseline is 92.55% (full sample).

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
XGBoost	0.9714	0.6462	94.02	56.75	82.92	0.6738	15s
CatBoost	0.9714	0.6456	94.02	56.74	82.87	0.6736	1m 14s
Random Forest	0.9712	0.6446	94.01	56.74	82.71	0.6731	1m 21s
H2O GBM	0.9710	0.6420	93.98	56.57	82.52	0.6713	31s
Lasso	0.9703	0.6338	93.94	56.42	81.94	0.6682	6m 25s
Elastic Net	0.9703	0.6338	93.94	56.42	81.93	0.6682	7m 11s
Logistic Regression	0.9703	0.6337	93.94	56.43	82.02	0.6686	1m 00s
Ridge	0.9701	0.6320	93.57	54.31	86.57	0.6675	4m 35s
Decision Tree	0.9680	0.5941	93.43	53.74	84.86	0.6581	6s
Naive Bayes	0.9189	0.3981	89.59	40.01	79.49	0.5323	1m 53s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	34,160.49	57.79
2	Established business owner	12,936.67	21.88
3	Self-efficacy	2,129.17	3.60
4	Knowing entrepreneurs	1,894.18	3.20
5	Opportunity perception	1,477.11	2.50
6	Self-efficacy (missing indicator)	855.58	1.45
7	High status (missing indicator)	568.50	0.96
8	Business angel	563.71	0.95

Share is the percentage of total forest impurity importance across all 25 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Source: Completed Objective 1 batch, 27 September 2026; 02_obj1_global_iv_OME. Scores and importance are taken directly from saved CSV outputs.